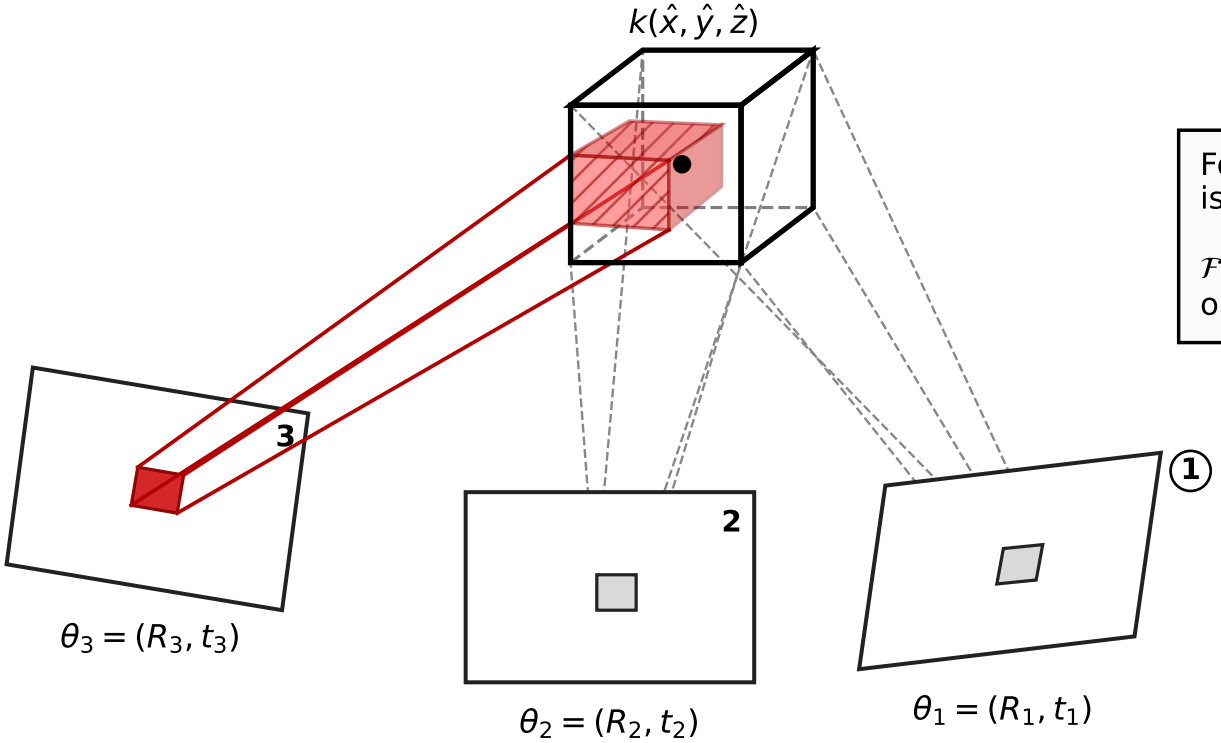

For example, the red volume is $S_k = \mathcal{F}_k \cap \mathcal{K}((R_3, t_3), W)$.

$\mathcal{F}_k$ is the original volume obtained by image 1 and 2.